GRATICULE INSTRUMENTS LTD

Applied AI research and production

Development Cost Statement

Applied AI platform development: model engineering, evaluation, interface and deployment

Statement of the effort, infrastructure and third-party consumption applied to the design, build, verification and evaluation of the platform, covering the period to 15 October 2026.

Basis: Commercial rate card

£54,850	£1,478	66	£56,328
Engineering effort	Infrastructure, data and API consumption	Engineer days	Total cost of development

Basis of preparation

Figures shown as MEASURED are verified against the delivered repository and are reproducible: code volume, artefact sizes, evaluation corpus size and verification results.

Day rates reflect senior UK contract market rates for the disciplines engaged.

Effort in days is derived from the delivered scope and should be reconciled to timesheets before external issue.

Third-party consumption is an estimate and should be replaced with provider billing data.

The entity is in the process of incorporation. No VAT is chargeable at this time and this statement does not constitute a demand for payment.

1. Scope delivered

All figures in this section are MEASURED from the repository.

1.1 Production code

Component	Modules	Lines
Model, inference and API layer (Python)	25	2,388
Interface layer (JavaScript, JSX, CSS)	16	1,438
Total	**41**	**3,826**

1.2 Trained artefacts and deployment assets

Artefact	Size	Status
model.pth (ResNet50 backbone, 3-class head)	94.36 MB	Delivered
LungDisease_Classifier.pth (LC25000CNN, 4,750,885 parameters)	19.01 MB	In service
lung_classifier.onnx (browser inference graph)	19.01 MB	In service
WebAssembly inference runtime, vendored and pinned	14.24 MB	In service
Technical and funding documentation	0.03 MB	Delivered

1.3 Verification and evaluation

Activity	Result	Basis
Numerical parity, PyTorch against ONNX Runtime	3.8e-06 maximum absolute difference	Measured
Labelled evaluation corpus	120 images, 40 per class	Measured
Label mapping determination	All 6 permutations evaluated, best identified	Measured
End-to-end smoke test	6 of 6 checks passed	Measured
Three-class accuracy on held-out data	Not yet established	Outstanding

The evaluation corpus is drawn from the same public dataset used for training and is therefore not a held-out result. Establishing a held-out split is the first milestone of the funded programme.

2. Engineering effort applied

MODELLED. Effort is derived from the delivered scope in section 1 and should be reconciled to timesheets.

Role	Day rate	Days	Amount
Principal Machine Learning Engineer (technical co-founder)	£950	24	£22,800
Senior Software Engineer, front end	£750	15	£11,250
Research Scientist, evaluation and validation	£800	9	£7,200
DevOps and Release Engineer	£650	6	£3,900
Technical Author, documentation and regulatory preparation	£500	5	£2,500
Clinical and Regulatory Advisor (fractional)	£1,200	3	£3,600
Commercial Lead (fractional)	£900	4	£3,600
Total engineering effort		**66**	**£54,850**

2.1 Effort distribution across the delivered scope

Workstream	Description	Days (full basis)
Discovery and architecture	Problem definition, data strategy, architecture and deployment model selection.	6
Model integration	Checkpoint reconstruction, architecture matching, inference layer, ONNX export.	10
Front end	Clinical dashboard, upload pipeline, results panel and explainability viewer.	12
Explainability	Occlusion sensitivity implementation to replace unavailable Grad-CAM.	5
Evaluation	Labelled corpus assembly, label permutation study, accuracy measurement.	9
Verification	Numerical parity testing, smoke tests, defect resolution.	7
Deployment	Static build, hosting configuration, release pipeline, runtime vendoring.	6
Documentation	Cost statement, funder deck, hosting assessment, technical notes.	6
Governance and commercial	Funder strategy, governance requirements, IP and entity framing.	5
Total		**66**

The workstream split above describes the scope as delivered across a full engagement, and totals 66 engineer days on that basis. Section 2 records the effort actually charged in this statement, which differs between scenarios because the rate basis differs. The scope delivered is the same in both.

3. Infrastructure, data and API consumption

MODELLED. Replace quantities with provider billing data before external issue.

Item	Quantity	Rate	Amount
Agentic development and code generation API consumption	28.5 M tokens	£2.10 / M	£60
Model training and fine-tuning compute	24 GPU hours	£2.40 / hr	£58
Evaluation and validation compute	60 CPU hours	£0.12 / hr	£7
Dataset acquisition, licensing and transfer	1 lot	£240	£240
Object storage and backup for artefacts	12 months	£14 / month	£168
Application hosting and content delivery	12 months	£20 / month	£240
Development tooling and design library licences	1 lot	£640	£640
Domain, TLS and identity services	1 lot	£65	£65
Total infrastructure and consumption			**£1,478**

Planned allocations not yet drawn, including any supercomputing allocation referred to in the funding strategy, are excluded and will be reported against the period in which they are consumed.

Invoice

INV-2026-0001 | Date: 15 October 2026

From

GRATICULE INSTRUMENTS LTD

Reference

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Description	Qty	Unit	Amount
Design, build, verification and evaluation of the applied AI platform, delivered as a working prototype with browser inference, evaluation harness and technical documentation (lot).	1	£54,850	£54,850
Infrastructure, dataset acquisition and third-party API consumption incurred in delivery (lot).	1	£1,478	£1,478
Subtotal			**£56,328**
VAT (not applicable, entity not yet VAT registered)			£0
Total due			**£56,328**

Purpose of this statement

This schedule records development expenditure incurred on the platform to date. It is prepared for internal capitalisation and funder disclosure, and is not a demand for payment.

Prepared by the technical function of GRATICULE INSTRUMENTS LTD. Figures marked MEASURED are reproducible from the delivered repository. Figures marked MODELLED are a planning schedule and are subject to confirmation.